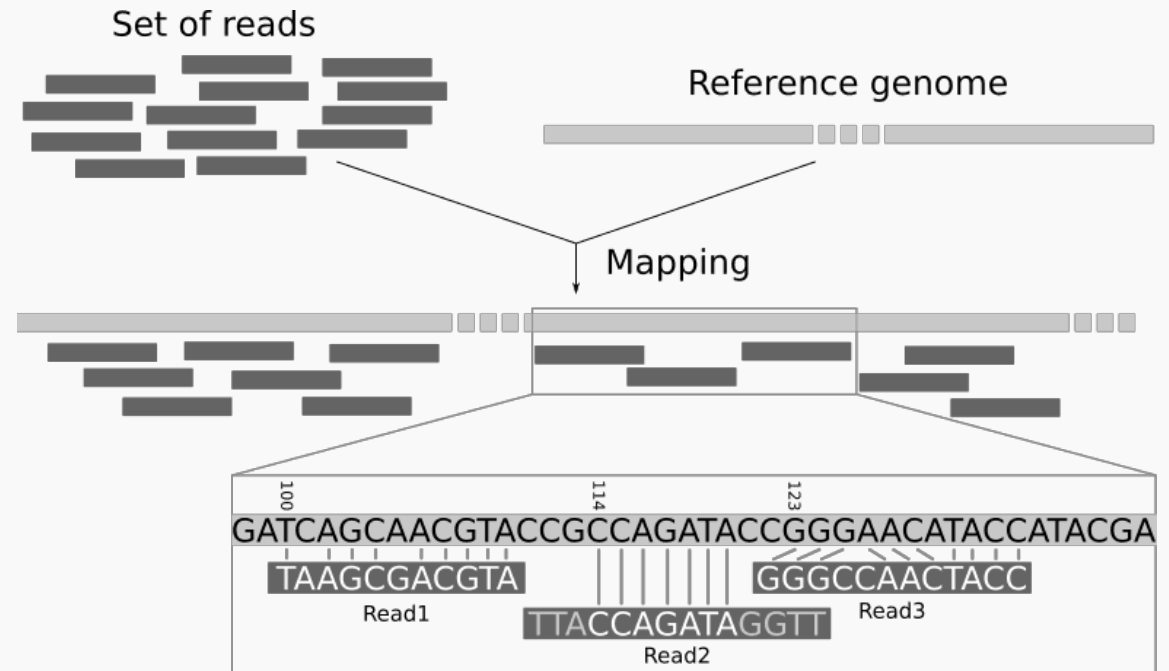


2. Mapping short-read data to a reference genome

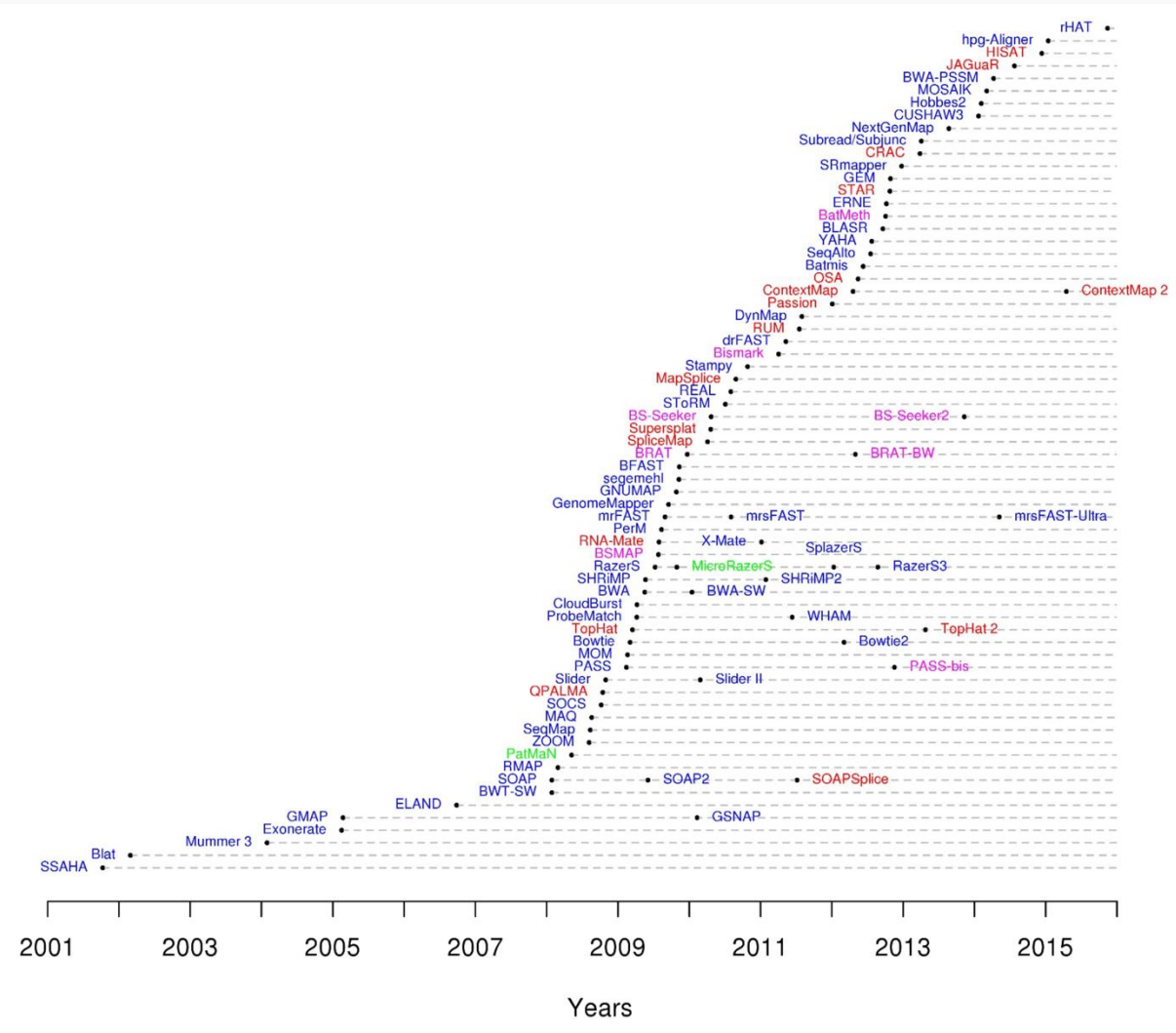
What is read mapping?

The process of finding where sequencing read most likely originated within a reference genome

<https://training.galaxyproject.org/training-material/topics/sequence-analysis/tutorials/mapping/tutorial.html>



More than 90 mappers to choose from!



<https://www.ecseq.com/support/ngs/what-is-the-best-ngs-alignment-software>

BWA-MEM2

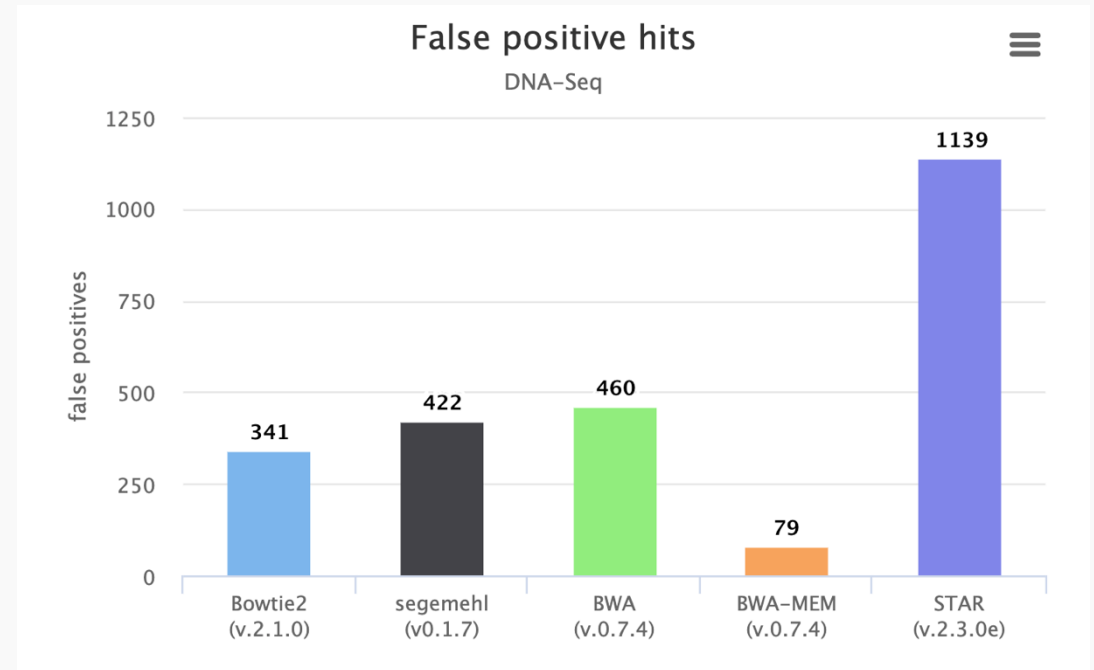
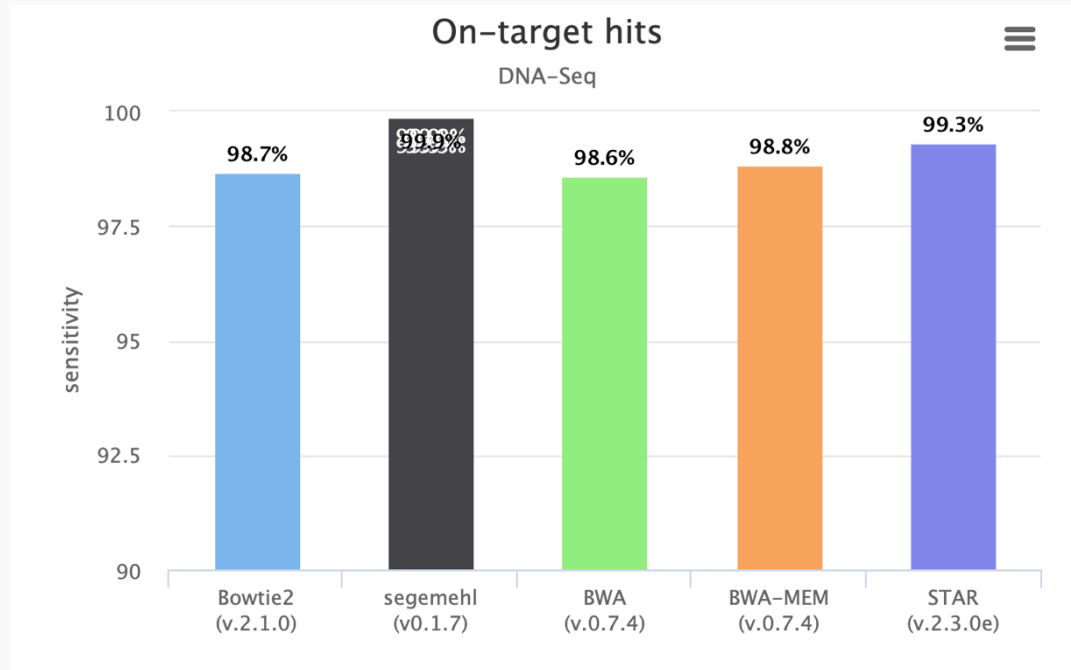
<https://github.com/bwa-mem2/bwa-mem2>

- Leverages the Burrows-Wheeler transform
- Very widely used, efficient, and open source
- The **Burrows–Wheeler transform (BWT)** makes short-read mapping efficient by reorganizing the reference genome into an index that can be **searched rapidly**.

https://en.wikipedia.org/wiki/Burrows%E2%80%93Wheeler_transform

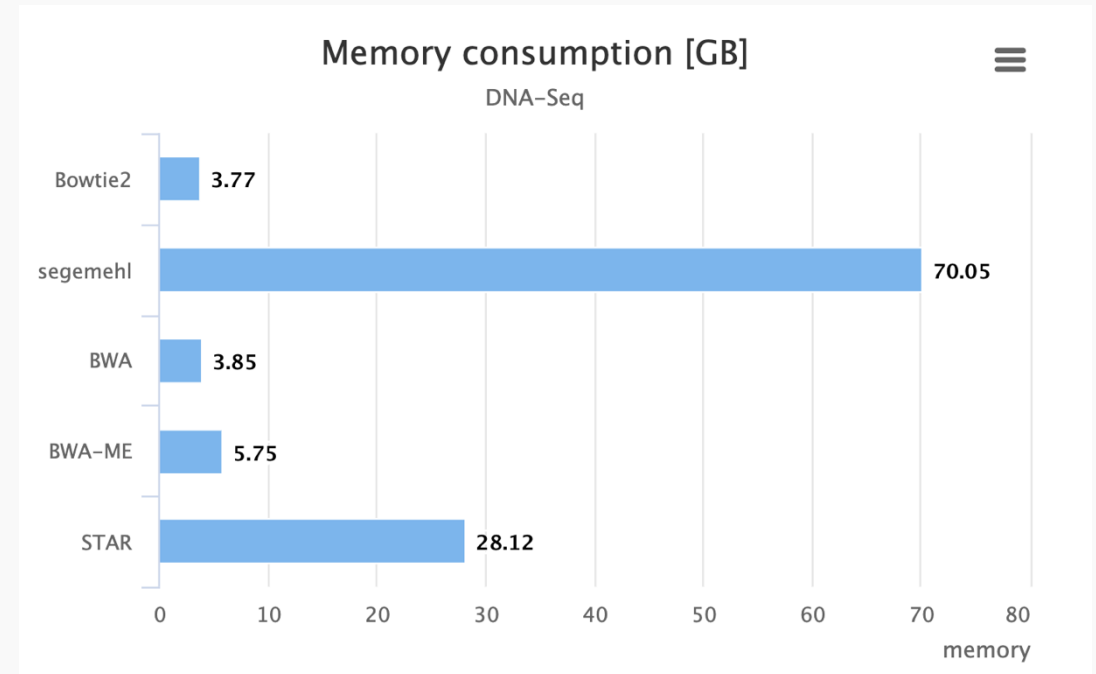
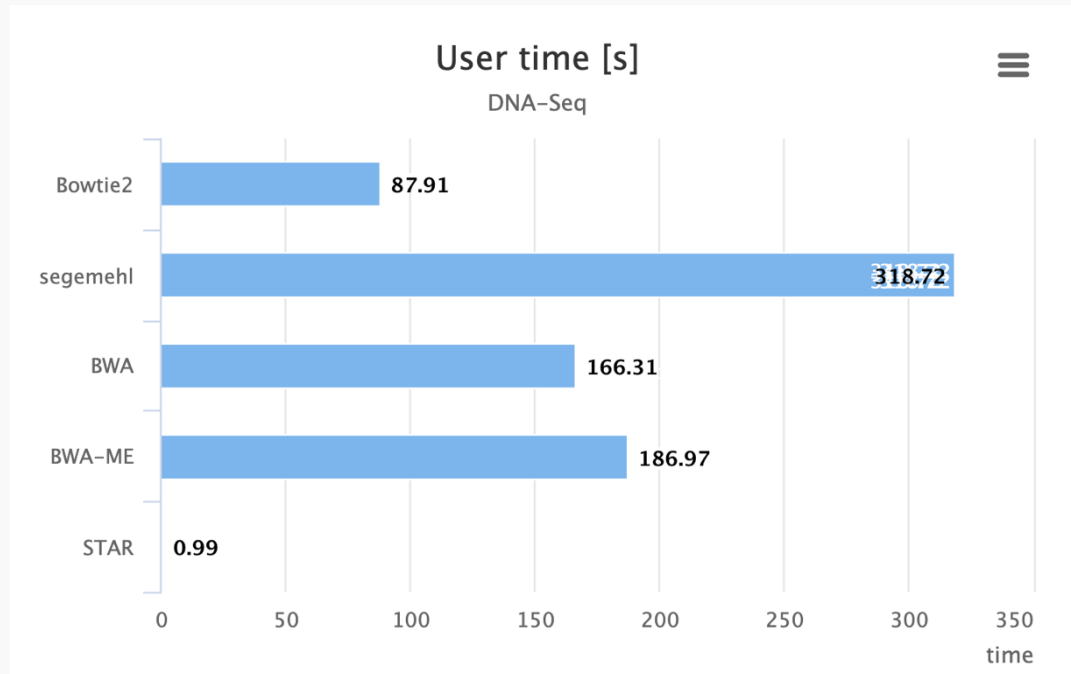
<https://www.youtube.com/watch?v=4n7NPk5lwbl>

BWA-MEM2: performance



<https://www.ecseq.com/support/benchmark>

BWA-MEM2: performance



<https://www.ecseq.com/support/benchmark>

SAM and BAM files

- **SAM (Sequence Alignment/Map)** is a human-readable, tab-delimited text file that records how sequencing reads align to a reference genome.
- contains an optional header followed by one line for each read alignment, including its genomic position, mapping quality, CIGAR string, sequence and quality scores.

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@HD VN:1.5 SO:coordinate											Header section
@SQ SN:ref LN:45											
r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*	Alignment section
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*	
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;	
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*	
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;	
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1	

Optional fields in the format of TAG:TYPE:VALUE

QUAL: read quality; * meaning such information is not available

SEQ: read sequence

TLEN: the number of bases covered by the reads from the same fragment. Plus/minus means the current read is the leftmost/rightmost read. E.g. compare first and last lines.

PNEXT: Position of the primary alignment of the NEXT read in the template. Set as 0 when the information is unavailable. It corresponds to POS column.

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CIGAR: summary of alignment, e.g. insertion, deletion

MAPQ: mapping quality

POS: 1-based position

RNAME: reference sequence name, e.g. chromosome/transcript id

FLAG: indicates alignment information about the read, e.g. paired, aligned, etc.

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r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*
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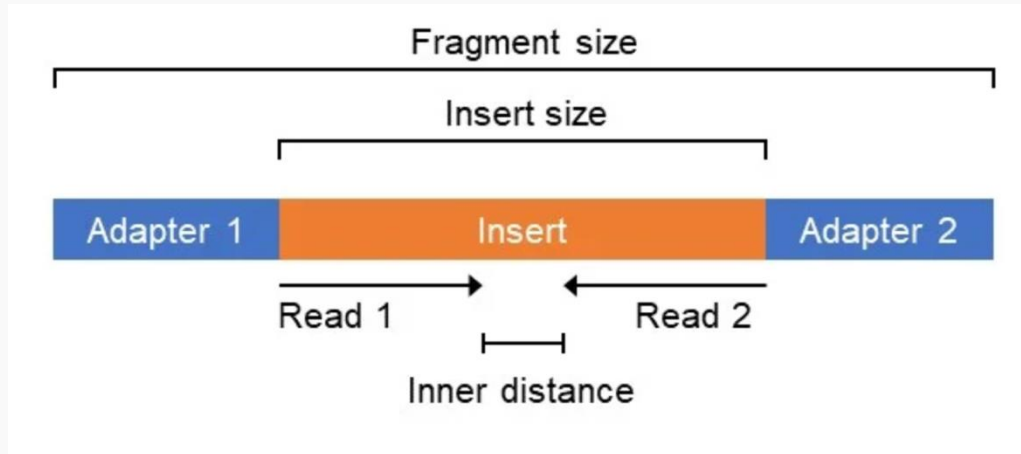
BAM (Binary Alignment/Map)

contains the same information but stored in a compressed binary format.

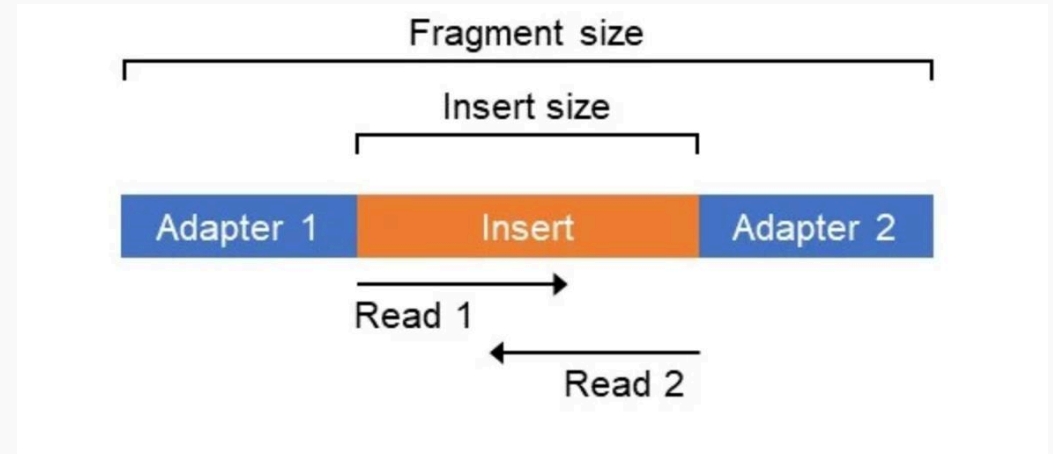
BAM files are smaller, faster for software to process and can be indexed, allowing programs to retrieve reads from a particular genomic region without reading the entire file.

Most Illumina sequences are now paired-end

Non-overlapping PE



Overlapping PE

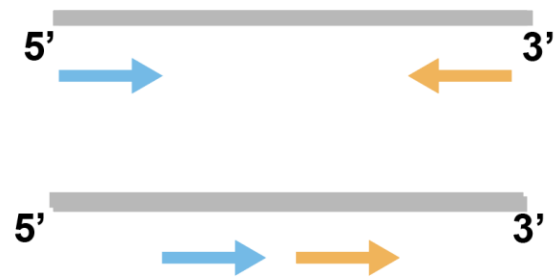


Paired reads allow us to assess mapping

Properly Paired Reads (PPR)



Improperly Paired Reads (IPR)



Mates map on different scaffolds



One of the mates does not map

